Predictive Maintenance – Fluid Pump Intervention

Standard Operating Procedure (SOP)

PURPOSE

This Standard Operating Procedure (SOP) defines the mandatory process for responding to predictive maintenance alerts related to Industrial Fluid Pumps (PUMP_1).

The purpose of this SOP is to ensure safe, consistent, and auditable intervention to prevent catastrophic pump failure caused by seal degradation or bearing wear.

SCOPE AND APPLICABILITY

This SOP applies to all maintenance personnel performing predictive maintenance interventions on Industrial Fluid Pumps (PUMP_1) located in designated Mechanical Rooms.

This procedure must be followed when interventions are triggered by the AI Engine and delivered through the Smart Action Pack. The SOP applies to all operational, maintenance, and engineering teams responsible for pump reliability.

ROLES AND RESPONSIBILITIES

- **Maintenance Technician** – Responsible for executing all steps in this SOP and ensuring safety procedures are followed.
- **Site Supervisor or Lead Engineer** – Responsible for verifying task completion, validating measurements, and approving work order closure.
- **Reliability Engineer** – Responsible for setting alert thresholds, reviewing recurring issues, and improving predictive logic.

OVERVIEW

This SOP must be followed when a predictive maintenance alert is generated indicating potential seal or bearing degradation in an Industrial Fluid Pump (PUMP_1).

The procedure defines the mandatory actions required to stabilise operating conditions and prevent asset failure.

TRIGGER CONDITIONS (ENTRY CRITERIA)

This SOP is initiated when all the following conditions are met:

- Pump discharge pressure fluctuates at or around **1.4 bar** for a sustained period exceeding X minutes.
- Vibration levels exceed **1.8 mm/s** or show unstable oscillation patterns.
- A Smart Action Pack notification is issued by the AI Engine.

HEALTH & SAFETY RULES

- Standard PPE must be worn at all times: impact-resistant gloves, steel-toe boots, and safety glasses.
- If ambient noise levels in the mechanical room exceed 85 dB, hearing protection is mandatory.
- A spill containment kit must be staged nearby before any work begins.

REQUIRED TOOLS

- Laser shaft alignment tool.
- Vibration analyzer pen for secondary manual verification.
- Torque wrench and adjustable wrenches.
- Asset-specific replacement mechanical seal kit.

STEP-BY-STEP EXECUTION PROCESS

1. **Preparation:** Acknowledge the dispatch from the Smart Action Pack and review the predictive timeline generated by the AI Engine.
2. **Locating the Asset:** Find PUMP_1 inside the designated Mechanical Room.
3. **Initial Inspection:** Conduct a visual check of the baseplate and pump casing to identify any loose mounting hardware or visible fluid leaks.
4. **Power Isolation:** Apply Lockout/Tagout (LOTO) protocols to the breaker at the pump's motor control center (MCC).
5. **Fluid Isolation:** Isolate the pump from the active line by closing both the upstream and downstream isolation valves.
6. **Component Check:** Inspect the pump coupling to verify if there is excessive wear or if the elastomeric insert has degraded.
7. **Shaft Alignment:** Attach the laser alignment tool to the pump and motor shafts. Correct any parallel or angular misalignment to successfully lower the vibration levels.
8. **System Restart:** Once resolved, open the isolation valves, remove the LOTO equipment, and start the pump.
9. **Observation:** Ensure the pressure stabilizes at its operational setpoint of approximately **2.5 bar** . Confirm that the vibration levels have fallen safely below the original alert threshold.

EXIT CRITERIA

This SOP may be considered complete only when all of the following conditions are met:

- Pump operating pressure has stabilised at approximately **2.5 bar** .
- Vibration levels are verified to be below the alert threshold.
- The pump has operated continuously for a minimum observation period without fault.
- All required evidence has been captured and uploaded.

EXCEPTION HANDLING AND ESCALATION

If pressure or vibration levels do not stabilise after corrective actions:

- Do not return the pump to service.
- Maintain lockout conditions.
- Escalate immediately to the Site Supervisor or Reliability Engineer.

If a fluid leak, excessive vibration, or unsafe condition is observed at any time, work must stop immediately and site safety procedures must be followed.

POST-JOB AUDIT AND EVIDENCE REQUIREMENTS

The following evidence is mandatory and must be uploaded to the D365 work order record:

- Final laser shaft alignment report.
- Post-intervention vibration measurement.
- Confirmation of pressure stabilisation.

All checklist items must be completed in the M365 Copilot interface. Work orders must not be closed until all evidence is verified and approved.

SOP GOVERNANCE

- **SOP Owner:** Reliability Engineering Team
- **Version:** 1.0
- **Approval Authority:** Head of Maintenance
- **Effective Date:** [Insert Date]
- **Review Cycle:** Annual or upon system change